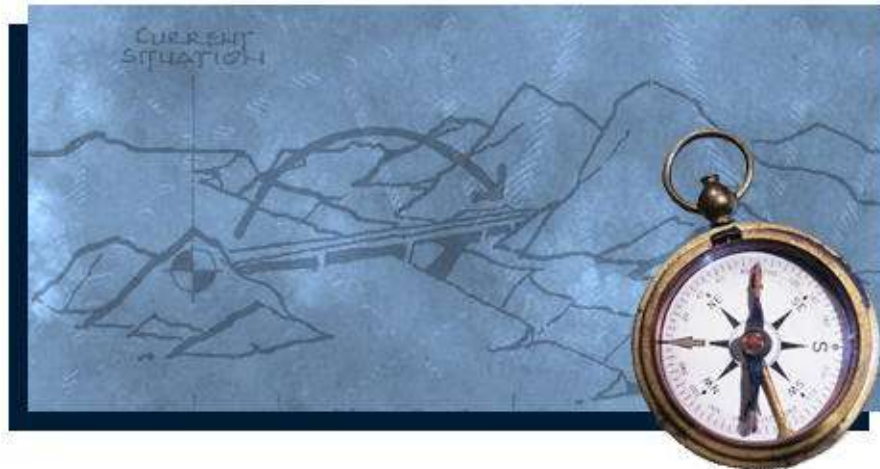




PRO Development™ Technical Overview

January 2010



Model and Theory

PRO Development™ is a career intelligence system founded on the belief that people will be fulfilled and productive when placed in a role that simultaneously taps their *motivations (drivers)*, *competencies (abilities)*, and *styles (personality)*. Traditional career counseling instruments have focused on one or two of these dimensions. Discerning that one has the “personality” to make a good attorney does little, however, to ensure that he/she has either the ability or motivation to be successful in this role. Such a limited one-dimensional approach does not even begin to address the type of law that an individual might be best suited for, let alone, the type of organization that would allow his/her career to flourish. Only a system that considers and *converges* on all three dimensions – motivations, competencies, and style (MCS) – will be able yield career choices that appeal to and play to an individual’s strengths and sense of purpose.

The concept of convergence is not a new one, and is fully discussed in the writings of Savickas [Savickas, Mark L. and Lent, Robert W. (1994). *Convergence in Career Development Theories*. Palo Alto, CA: CPP Books.]. During the early 1990’s, Savickas and other leading career theorists were calling for a new, comprehensive, fully convergent model to articulate, measure and integrate competencies, motivations and personality factors. Obik (a Princeton-based assessment firm) developed such a system, after a thorough review of hundreds of instruments and career counseling and development programs. Based upon these works, Obik developed a system to measure the MCS dimensions. In the beginning, existing tools with demonstrated reliability and validity were selected, for example, the Enneagram, DISC, Life Styles Inventory, and Organizational Culture Inventory (see Appendix A for a more complete sample of instruments reviewed and utilized). Between 1992 and 2002, the Obik team developed and tested its own specific set of MCS tools, and created the algorithms necessary to create highly individualized convergent models. One of them specifically developed was the PRO Development™ tool, specific for analyzing and providing for guidance for the professional growth and development of leadership individuals and teams.

In 2003, TAI secured the exclusive rights for PRO Development™ (PRO-D). TAI’s PRO Development™ is not a “test” – it is an experiential process that helps individuals understand how their unique set of motivations, competencies, and style/personality can be simultaneously expressed and experienced in a number of different career choices and organizational relationships. Just as blue, green, and red can be combined to create 12 million different colors, the sensitivity and specificity of the PRO-D system allows for thousands of possible career choices – far more individualized than traditional vocational instruments.

In order to organize MCS around a cohesive framework, PRO-D utilizes a set of nine ‘archetypes’ called Career Value Archetypes (CVA’s). These archetypes were identified after an exhaustive review of the literature and consideration of multiple models. Taken individually, the archetypes describe elements of behaviors, attitudes, and thoughts. However, when combined, they describe types and classes of people and constellations of behaviors, attitudes, and thoughts. (See Appendix B for a list of the nine archetypes and Appendix C for a comparison of well-established theories.) These archetypes allow us to determine how MCS manifests itself in specific roles and, ultimately, how motivations, competencies, and style converge to identify roles that consider each of these factors simultaneously. The convergence process is more than “a sum of the parts” and the algorithms vary depending on the product selected and needs of the individual client.

PRO Development™

Each individual takes an on-line, web-based survey. PRO-D is built using a modular approach, and consists of a grouping of several psychometric tools. The survey is not timed, and is self-administered. The individual is advised that there are no right or wrong answers, and they can start and stop as frequently as needed and then can come back and complete the remainder of their survey. However, we do ask that the participant complete the current tool in its entirety before stopping.

Upon completion of the survey, an interpretive report and graphical map (a visual representation of their results in a synoptical format) are automatically generated and available for delivery to a TAI certified PRO-D facilitator.

Validity and Reliability

The nature, purpose and use of PRO-D profoundly shape the kind of validity and reliability strategies that are appropriate. Test-retest approaches presume a single trait, factor and/or outcome. Due to the multi-dimensional factors of PRO-D, traditional test-retest approaches are not appropriate and useful for PRO-D system.

The power of a tool like PRO-D does not rest alone in the assessment. Rather than measuring a desired gain on a single variable, relevant in all cases, or on a collection of separate variables, the validity and reliability strategies take into account the fact that PRO-D is designed to be used as a component of a personal discovery and development process. TAI and those certified as PRO Development consultants support a holistic development process in a highly customized intervention over time, rather than the reliance of a one time, generic event.

The personal discovery process is highly individualistic and plays out in the context of a dynamic, on-going interaction of factors (motivations, competencies and (personality) style dimensions) within and intrinsic to the individual, while at the same time interacting with numerous powerful, and elusive external variables such as external events and input from significant others. In the personal discovery process, each individual has a unique set of desired outcomes – at the time of the intervention – which shape the way effectiveness of the intervention must be measured. An individual's specific goals may change over time and the individual will make choices, which set new "variables" in motion.

An analogue to the supported personal discovery process is an organizational change project. The criteria for determining the effectiveness of such an intervention will be set by the organization and will likely consist of a number of desired outcomes, defined and weighted by the organization. The effectiveness of such an intervention will try to take into account a number of exogenous factors (such as the economic state of the market place) that are difficult to predict and track. Finally, in most cases, the organization itself will play a significant, on-going role in determining success or failure.

Given the nature, purpose and use of PRO-D, the validity and reliability research has several primary components:

- **Reliability** - demonstrating inter-item consistency and appropriate test-retest consistency (allowing for the fact that individuals change over time, especially in the motivational area).
- **Tool-component validity** - showing that the specific components ("tools") of the PRO-D system do measure, with independence and robustness, what they purport to measure. Obik conducted research to show Face, Content, Construct, and Concurrent/Criterion Validity. Ongoing validity studies continue.
- **Predictive validity** (Positive Impact/Effectiveness) - showing that PRO-D, in the context of a personal discovery process, makes a significant, positive impact on those who engage in the process. "Predictive Validity" needs to be defined in a way that takes into account the highly individualistic and interactive nature of the PRO-D process. Obik conducted longitudinal impact surveys that demonstrate favorable results.

Psychometric Properties

Reliability

Internal Consistency

In order to test whether the individual scales that compose the PRO-D battery are internally consistent, coefficient alpha was computed on several tools by archetype (CVA). The vast majority of scales demonstrated solid internal reliability with alpha values at or above .65. (Note: Analyses were computed on a set of over 500 clients who completed *PRO Development*™ v3.2.)

- Alpha Reliability Coefficients for Abilities Archetypes – alpha values ranged from .56 to .90 with an average of .78
- Alpha Reliability Coefficients for Style/Personality Archetypes – alpha values ranged from .64 to .78 with an average of .71
- Alpha Reliability Coefficients for Role Archetypes – alpha values ranged from .52 to .80 with an average of .70
- Alpha Reliability Coefficients for Organization Archetypes – alpha values ranged from .58 to .82 with an average of .72

Validity

Face Validity

Does the tool appear to measure what it is supposed to measure? For example, if we are assessing one's penchant for becoming a chef, would it make sense to ask about his/her favorite color or number? This form of validity is given little weight in evaluating an instrument's psychometric soundness, but is important for maximizing a client's belief that the instrument is valid and thus enhances his/her efforts to fully engage and participate in the process.

TAI collects satisfaction ratings from every client that goes through our program. We use this information to evaluate and improve the impact of our products, services, and programs. The following customer satisfaction data is based upon clients who completed the PRO Development™ product. In Appendix D, we have collected a number of testimonials from clients about their experience across a number of products. Based on this feedback, face validity has clearly been established.

Customer Satisfaction Ratings

1. On a scale from A, B, C, D, F, 100% gave the report a grade of "A" or "B."
2. 100% responded that the program was useful.
3. 83% of participants found the PRO Development™ to be superior to other tools that they had taken.
4. 99% gave PRO-Development™ an Overall Satisfaction Rating.
5. 97% responded that they would recommend the process to others.

Content Validity

Do the items of a test (usually of the achievement variety) adequately cover all of the aspects of the content area in terms of breadth and specificity? Item consideration and selection underwent an extensive process and involved many experts in the areas of education, psychology, surveying, testing, and career counseling. Most influential was the review conducted by the Princeton Testing Group.

The Princeton Testing Group

Name	Title	Organization
Dr. Tom Bachhuber	President	Bachhuber and Associates
Victoria I. Ball	Dir. of Career Planning & Placement	Brown University
Dr. Eric R. Braverman	President	Princeton Assoc. for Total
Henry P. Bristol, II	Principal	Pikaart-Bristol Associates
David H. Collins	President	Obik, Inc., TranZitions, Inc.
George H. Gallup, Jr.	Co-Chairman	Gallup Organization
Dr. Mary J. Gregerson	Professor of Psychology	George Washington Univ.
John G. Patterson	English Teacher, Curriculum Consultant	St. Paul's School
Kathy S. Worgul	Independent Counselor	University of Texas

This group in particular met several times to review and revise individual items, as well as review and update program materials and applications. Thus, we are confident that our items sufficiently measure the constructs they are intended to.

Construct/Convergent Validity

In order to ensure construct validity, we correlated our personality components with DISC – a well-known and highly regarded instrument with demonstrated psychometric properties. By correlating the major scales of Dominance, Influence, Steadiness, and Compliance with our Personality tool, were able to support convergent validity.

Correlations Between DISC Scales and PRO-D Archetypes

DISC Scales	PRO-D Archetype	Value
"Dominance"	"Entrepreneurial Challenge" (Driving)	($r=.53$, $p<.001$)
"Dominance"	"Natural Appreciation" (Observing)	($r=-.35$, $p<.001$)
"Influence"	"Motivational Energy" (Expressing)	($r=.58$, $p<.001$)
"Influence"	"Natural Appreciation" (Observing)	($r=-.38$, $p<.001$)
"Steadiness"	"Natural Appreciation" (Observing)	($r=.44$, $p<.001$)
"Steadiness"	Entrepreneurial Challenge" (Driving)	($r=-.51$, $p<.001$).
"Compliance"	"Natural Appreciation" (Observing)	($r=.41$, $p<.001$)
"Compliance"	"Motivational Energy" (Expressing)	($r=-.45$, $p<.001$)

In addition, we conducted a canonical correlation between the PRO-D archetypes and factors of the Myers-Briggs Type Indicator (MBTI). Results indicate that the archetypes account for 42% of the variance in the MBTI factors. In contrast, the MBTI accounted for only 27.5% of the variance in the archetypes. These findings suggest that the archetypes capture substantially richer personality profile than does the MBTI.

Finally, we correlated all nine archetypes with one another to ensure that they were distinct from one another. Correlations between the archetypes averaged .43 with only two correlations reaching the .71 level – suggesting at most 50% shared variance between any two archetypes.

Concurrent/Criterion Validity

In order to demonstrate that our tools can accurately assess an individual's MCS, we identified three different groups of clients who had gone through the process: Bonner Scholars, The Chase Manhattan Bank, and SetFocus. Additional research showed that the PRO-D process accurately identified those individuals as good fits for their roles. Future research will be conducted to further validity studies.

Positive Impact/Effectiveness ("Predictive Validity")

Long Term Impact

Most organizations focus on a client's satisfaction in the short-term – immediately after the program. At TAI, our product and process is intended to impact long-term change, and we are actively concerned that our work continues to satisfy and improve the life of the client long after the initial experience. Various surveys seek to measure this. One recent survey of 41 clients (70% male, mean age 41 years) who completed our program an average of three years ago (range 6-74 months) suggests that the experience is both positive and powerful over time.

Appendix A: Instruments Reviewed

- Johnson O’Conner and Ball Aptitude Batteries - 19 and 12-scale aptitude tests
- Participant evaluation forms from TranZitions’ career workshops
- 10 surveys conducted through Rutgers University, Denison, Middlebury, Ohio State, Kenyon College, and Brown University - over 3,000 participants
- Gallup Personal Survey techniques used by Gallup Poll and Gallup Institute
- Marston 4-factor psychological model for behavioral analysis
- Jung 4-factor psychological model for psychological analysis
- Cassell’s 16 Personality Factor model for assessment
- Ed Schein’s, MIT, model for 8 Career Anchors
- Holland and Strong-Campbell Interest Tools
- Missions model for spiritual development
- Wonderlic model for intellectual assessment
- 14 core Aptitude discoveries - Harvard
- Development of Individual Developmental Training curriculum -- multi-company project
- In-depth study of organizational cultures - Council on Economic Priorities
- 3 years active use of alternate career tools for comparisons to OI tools
- Development of key non-profit job referral network based at Harvard’s Kennedy School of Government
- Factor analysis of the NEO Five Factor Model for psychological assessment
- Matrix methodology for solving three complex equations simultaneously

Appendix B: A Description of the Nine Archetypes

Influencing (Societal Change)	Influencing is concerned with effecting change to things around it - society, rules, policies, and government. Strongly held points of views, opinions and beliefs foster deep commitments and desire to influence others.
Analyzing (Theoretical Discovery)	Analyzing is concerned with analyzing, determining underlying reasons and facts. It becomes central to "dig deep" and discover root causes for occurrences so as to answer questions and explain why and how something occurred.
Synthesizing (Strategic Decisions)	Synthesizing is concerned with integrating findings into a collective understanding. Looking at a range of issues, finding patterns, and understanding impacts of alternative scenarios enables effect planning and decision-making.
Helping (Human Development)	Helping is concerned with reaching out and assisting others with a need they have. Compassion and empathy create the impulse and commitment to help and resolve issues for others.
Driving (Entrepreneurial Challenge)	Driving is concerned with generating sufficient energy to create a desired outcome or effect. Overcoming challenge and competing against the elements creates its own rewards of winning and achieving.
Producing (Production Efficiency)	Producing is concerned with action, order, and accomplishment. Building and hands-on work create tangible results, efficiency, and practical sense.
Creating (Artistic Creativity)	Creating is concerned with thinking new ideas and sharing visions through artistic means. Designing through innovation and interpretation creates new products and unique works of art.
Observing (Natural Appreciation)	Observing is concerned with keen listening and watching out for balance and ensuring people maintain a respect for life. A care for natural order necessitates observing to guard and protect our resources, peace, nature, and public spaces.
Expressing (Motivational Energy)	Expressing is concerned with motivation and showing enthusiasm so others can be entertained and enjoy certain moments in their lives. Exciting displays of expression enables people to enjoy a break from their routines.

Appendix C: Model Comparisons

Below are two tables that present the theoretical relationship between PRO-D's model and four well-known systems of career development. The first table compares PRO-D to Edgar Schein's Model known as Career Anchors, John Holland's Hexagonal Model of Vocational Personalities and Work Environments, and Anne Roe's Classification of Occupational Groups. As can be seen from the table, there are no significant gaps between the PRO-D Model and these other popular models of career development.

PRO-D Archetypes (CVA's)	Schein¹ Career Anchors	Holland² Hexagonal Model	Roe³ Occupational Groups
Influencing	Service/Dedication to a Cause	Social, Enterprising	General Culture
Analyzing	Technical/Functional Competence, Autonomy/Independence	Investigative, Realistic	Science, Technology
Synthesizing	General Managerial Competence, Technical/Functional Competence	Enterprising, Conventional, Realistic	Organization, Business Contact
Helping	Service/Dedication to a Cause	Social	Service
Driving	Entrepreneurial Creativity, Autonomy/Independence	Enterprising, Conventional	Business Contact
Producing	Technical/Functional Competence, General Managerial Competence, Security/Stability	Realistic, Conventional, Enterprising	Organization, Outdoor
Creating	Autonomy/Independence	Artistic	Arts & Entertainment
Observing	Security/Stability, Lifestyle	Realistic, Conventional	Outdoor
Expressing	Pure Challenge	Enterprising, Social	Business Contact, Arts & Entertainment

Theoretical Relationship Between the PRO-D Model and ACT's Work Keys Skills

Skills	PRO-Development™ Archetypes			
Applied Mathematics	Analyzing	Synthesizing	Producing	Creating
Applied Technology	Producing	Observing	Analyzing	Driving
Listening	Helping	Synthesizing	Expressing	Influencing
Locating Information	Synthesizing	Analyzing	Producing	Observing
Observation	Observing	Analyzing	Synthesizing	Influencing
Reading for Information	Analyzing	Producing	Synthesizing	Influencing
Teamwork	Helping	Producing	Observing	Expressing
Writing	Influencing	Producing	Observing	Creating

ACT's Work Keys Skills assessment is a measure of competencies required to perform work. It is a component of ACT's larger system for analyzing and describing the characteristics needed to perform specific jobs and occupations. Again, it appears that the two systems are integrated in both Models. However, it appears that the

skill areas measured in the Work Keys is more heavily concentrated in the analyzing, producing and synthesizing archetypes in PRO-D's Model.

Appendix D: Customer Testimonials

"It's scary how on target this was for me. Thanks!"

"You absolutely nailed me!...After 62 years it is nice to know I have spent the greater part of my career in absolutely what I do best. Maybe that is why I have been so happy and content with my career."

"As I was saying after the training today - I hope people are aware of how valuable the training and development profiles that PRO-D provided are. In comparison to other professional development workshops and career development counseling - the information and analysis included in the PRO-D Profile is more focused and ultimately more useful. To be given insight in how personal and professional strengths can be better applied within the "X" organization is very pro-active, motivating information. It is also very much reflective of the corporation's desire to retain talented employees. A critical insight provided by the report shows where discrepancies between competencies and interests exist, helping employees to make choices from the educational/developmental resources that exist here at "X" and move towards attaining professional goals and personal satisfaction. The way the profile is organized, its in-depth, individual structure provides a way to see yourself in the context of your work and workplace, which is invaluable. The report is a catalyst for positive change."

"I have recommended PRO-D to many of my friends and to most companies I have interviewed with. I am better off using the PRO-D profile than using other profile tools found in books or at the outplacement agency that was provided by my last employer."

"PRO-D helped me understand why my career choice was not working for me. I selected a career that matched my technical skills, but not the goals I wanted to achieve. This insight helped me evaluate several job offers I had using the proper criteria."

"Articulation and confirmation of what my "Gut" was telling me."

"That my desired work/career choice was what I wanted to do & didn't 20 years ago."

"At first I viewed this as a Myers Briggs' type of program — the difference in this program is that PRO-D is geared toward ensuring the "right people" are in the "right roles."